

AKHILSAI CHITTIPOLU

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EDUCATION

- **North Carolina State University - Master of Computer Science** - 4.00/4.00 Aug 2023 - May 2025
Coursework: Data Analysis & Algorithms, Software Engineering, Database Management Systems, Computer Networks, Neural Networks and Deep Learning, Object-Oriented Design and Development, Automated Learning and Data Analysis, Privacy in the Digital Age, Human-Computer Interaction.
- **VNRVJIET - Bachelor of Technology, Information Technology** - 9.00/10.00 Jun 2018 - May 2022

TECHNICAL SKILLS

- **Programming/ Scripting Languages:** Python, C++, Java, Javascript, C#, Ruby, C, R, Linux.
- **Web Technologies:** Angular, HTML, CSS, Bootstrap, ASP.NET, GIT, Ruby on Rails.
- **Databases and Tools:** MySQL, Microsoft SQL, Toad, Oracle Databases.
- **DataScience Platforms& Frameworks:** Machine Learning, Data Analytics, Tkinter, Pandas, NumPy.
- **Cloud Technologies:** AWS Cloud - Elastic Beanstalk, Redis, Lambda, RDS, SQS, CloudWatch.

WORK EXPERIENCE

IT Application Developer at State Street, Hyderabad, Telangana, India Jul 2022 - Aug 2023

- Developed a web application named the FAR (Fund and Asset Reporting) application, dedicated to collecting asset data for serviced clients at the fund level, ensuring data integrity and streamlined reporting processes.
- Enhanced the user interface (UI) of the FAR development website using ASP.NET, resulting in a 20% increase in customer satisfaction and a 15% decrease in end-user support inquiries. Delivered a more intuitive, user-centric experience, significantly enhancing client engagement and usability.
- Actively contributed in optimizing C# code to improve communication between the front-end and back-end databases, achieving a 30% boost in data access efficiency and enhancing overall application performance.
- Modified the web database of the FAR Database using Microsoft SQL, resulting in a 25% reduction in data retrieval times and a 10% improvement in overall database efficiency to better meet the project requirements. This involved adapting the database structure and data to align with the evolving needs of the application.
- Utilized tools such as Microsoft Visual Studio Code, Microsoft SQL Server Management Studio, and Git (source control) to optimize daily development tasks, enhancing code quality and version control efficiency.

Software Intern at State Street, Hyderabad, Telangana, India Jan 2022 - Jun 2022

- Gained extensive hands-on experience with Oracle E-Business Suite (EBS), PL/SQL development, and Oracle RAC databases, optimizing SQL queries using tools like Toad. Achieved a 15% boost in query efficiency and a 10% reduction in execution time, significantly enhancing database performance, scalability, and reliability.
- Proactively contributed to a high-performing CI/CD Kanban environment, driving impactful feature development and efficient bug resolution within an Agile Scrum framework to ensure continuous improvement.

PROJECTS

Machine Learning using K-Means Routing Mechanism (Malaria Parasites Classification)

- Developed an innovative Compositional Coding Capsule Neural Networks with K-Means Routing Mechanism for Malaria Parasites Classification, advancing the state-of-the-art in disease diagnosis.
- Trained the application on a dataset consisting of 27,558 thin blood smear images, comprising both infected and healthy erythrocytes, sourced from the NIH repository model development and accuracy enhancement.
- Our model outperformed other deep learning techniques, achieving an impressive 99.2% accuracy and 91.2% validation accuracy, demonstrating its exceptional predictive power and reliability in real-world scenarios.

Deep Learning using Transformer Encoder (Classification between Real vs. AI-generated Images)

- Developed a deepfake image classification model using TensorFlow, Keras, and Python, combining ResNet50 and VGG16 feature extraction with a Transformer encoder for real vs. AI-generated face detection.
- Engineered a hybrid architecture that integrates CNN-derived spatial features with Transformer blocks using multi-head attention and positional embeddings to capture global relationships in image embeddings.
- Evaluated model robustness by applying noise, blur, rotation, and brightness distortions to test data, and demonstrated strong generalization across unseen datasets for deepfake detection use cases.

Machine Learning / Deep Learning (Emotion Detection System)

- Devised a novel emotion detection system employing deep learning techniques to accurately recognize seven categories of human emotions through facial expressions, capable of real-time predictions.
- Trained the application on an extensive dataset, using a highly optimized Convolutional Neural Network and VGG 16 models, achieving remarkable training and validation accuracies of 97.3% and 95.7%, respectively.